GGR376 Final Project: Spatial Analysis of Diabetes in the Census Tracts across Toronto, Peel, and York Region

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1 INTRODUCTION & RESEARCH QUESTION

Diabetes is a major public health concern, and its prevalence is influenced by a range of social and environmental factors. Among these, education plays an important role in shaping health outcomes by influencing health awareness, lifestyle choices, and access to information. (Zajacova & Lawrence, 2018). Previous research has shown that individuals with higher levels of education are more likely to engage in healthier behaviours and better manage chronic conditions, while lower educational attainment is often associated with limited access to health knowledge and preventive care (Luo et al., 2025). In addition to education, factors such as income and access to recreational spaces have also been linked to health outcomes. Studies highlight how physical activity and access to green space contribute to improved health and lower mortality risks (Maas et al., 2006), while active daily routines can support glycemic control for individuals living with diabetes (Patience et al., 2023). These findings suggest that both socioeconomic and environmental conditions may influence the distribution of diabetes.Given these relationships, it is important to examine how education relates to diabetes prevalence within an urban context. This study focuses on census tracts within Toronto, Peel, and York regions and investigates whether differences in educational attainment are associated with variations in diabetes prevalence. It also explores whether diabetes prevalence exhibits spatial patterns across the study area.

2 DATA & STUDY AREA

The study focuses on census tracts within the City of Toronto, Peel Region, and York Region. Data will be primarily sourced from Statistics Canada for its accessibility, comprehensiveness, and quality. The primary health data and income data is sourced from the Canadian Community Health Survey administered by Statistics Canada. The dataset includes the raw number of people over 12 years old who have diabetes (Type I and II) as well as the total population surveyed in each census tract in the Greater Toronto Area for 2025. The other dataset for recreational spaces

is located in the Open Database of Recreational and Sport Facilities (ODRSF), also maintained by Statistics Canada. The database compiles recreational spaces sourced from provincial, territorial, and municipal governments across Canada and is up to date as of 2021. The main variables the team will be using in this dataset are the longitude, latitude, and type of recreational facility of each recreational space. The education data is also sourced from CHASS which originates from Statistics Canada Census (2021). The education data includes the percentage of those who earned no certificates, those who completed a high school education, and those who completed post-secondary education by census tract.

3 METHODOLOGY

All the data was extracted from various open data portals as mentioned in the Data section. The datasets were inspected–null values were removed, data types were validated, and semantic difference such as leading or trailing characters were removed to prevent any discrepancies and join multiple datasets via census tract in the further analysis. This study is composed of a combination of exploratory spatial data analysis (ESDA), regression analysis and spatial autocorrelation to determine the relationship between education (percentage with no certification, percentage with only high school completion, percentage completed post-secondary) and diabetes prevalence across Toronto, Peel and York regions. Before running any spatial autocorrelation analysis, a spatial weights matrix was built to define which census tracts count as neighbours to one another. Queen contiguity was used, meaning two tracts are considered neighbours if they share either an edge or a corner. This made more sense for this dataset than Rook contiguity, which only recognises shared edges. In the study area, there are plenty of cases where tracts only meet at a single corner point, and Rook would have missed those relationships entirely. Distance-based approaches like KNN were also considered, but the size difference between dense urban tracts in Toronto and the much larger suburban tracts in Peel and York made a consistent distance threshold impractical. The matrix was built using Queen from PySAL and row-standardised, so each tract's neighbours receive equal weights that sum to one. This is the standard approach for Moran's I and LISA to prevent tracts with more neighbours from having an outsized influence on the spatial lag. One tract was found to be a spatial island with no neighbours due to its geographic isolation and was dropped before rebuilding the matrix. The final weights matrix included 1,059 census tracts and was confirmed to be fully connected with no isolated components. Global Moran's I was calculated for each

variable to test whether similar values tend to cluster together across the study area or are spread randomly. Significance was tested using 999 permutations, and Moran scatterplots were produced to visualise the spatial relationships. Local Moran's I (LISA) was then used to pinpoint where those clusters actually appear on the map, labelling each tract as High-High, Low-Low, High-Low, Low-High, or non-significant at $p < 0.05$. Bivariate Moran's I was also run to test whether high or low diabetes in one tract tends to be spatially associated with high or low values of a predictor in neighbouring tracts. Both global and local versions were mapped for each predictor variable. To further understand the spatial patterns identified, a Gini coefficient and Lorenz curve were computed for each variable to measure how unevenly a variable is distributed across census tracts.

The spatial distribution maps were created to understand the spread and clustering of the dataset in the study area according to the Queen Contiguity spatial weights using K-means and Ward's Hierarchy methods. Moreover, an Ordinary Least Squares (OLS) regression model was carried out, along with scatterplots and correlation analysis to estimate the statistical relationship between the variables. Furthermore, the spatial regression models, including spatial error and spatial lag, were conducted to capture the spatial processes in the data and determine the relationships of nearby regions.

4 RESULTS

Figure 1 shows the study area, which includes Toronto, York, and Peel, while Figure 2 is a choropleth map of diabetes rates among persons aged 12 and older by census tract across these regional municipalities. The data drawn from the 2025 Canadian Community Health Survey. For each census tract, the rate was calculated by dividing the crude number of surveyed respondents with diabetes (ages 12+) by the total surveyed respondents in the same age group. Normalizing accounts for population size differences and allows comparisons across census tracts. Figure 2 also shows that Toronto contains far more census tracts than York and Peel, with several scattered hotspots and coldspots visible across the study area.

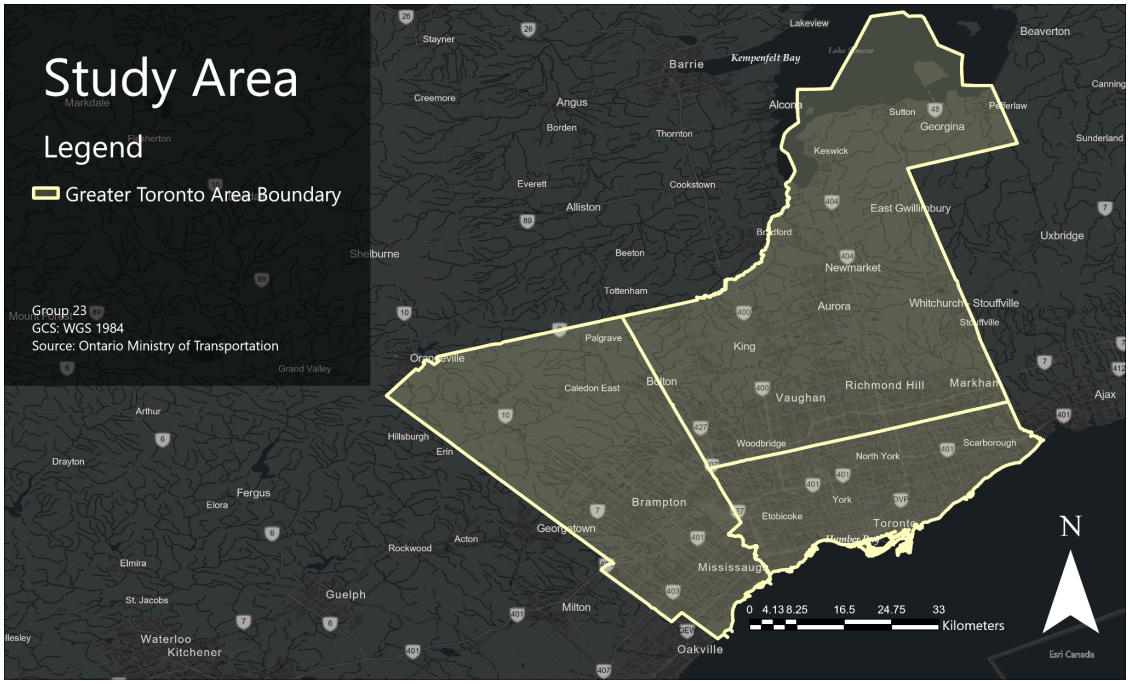


Figure 1 (above): The study area of the Peel, York, and Toronto Regional Municipalities.

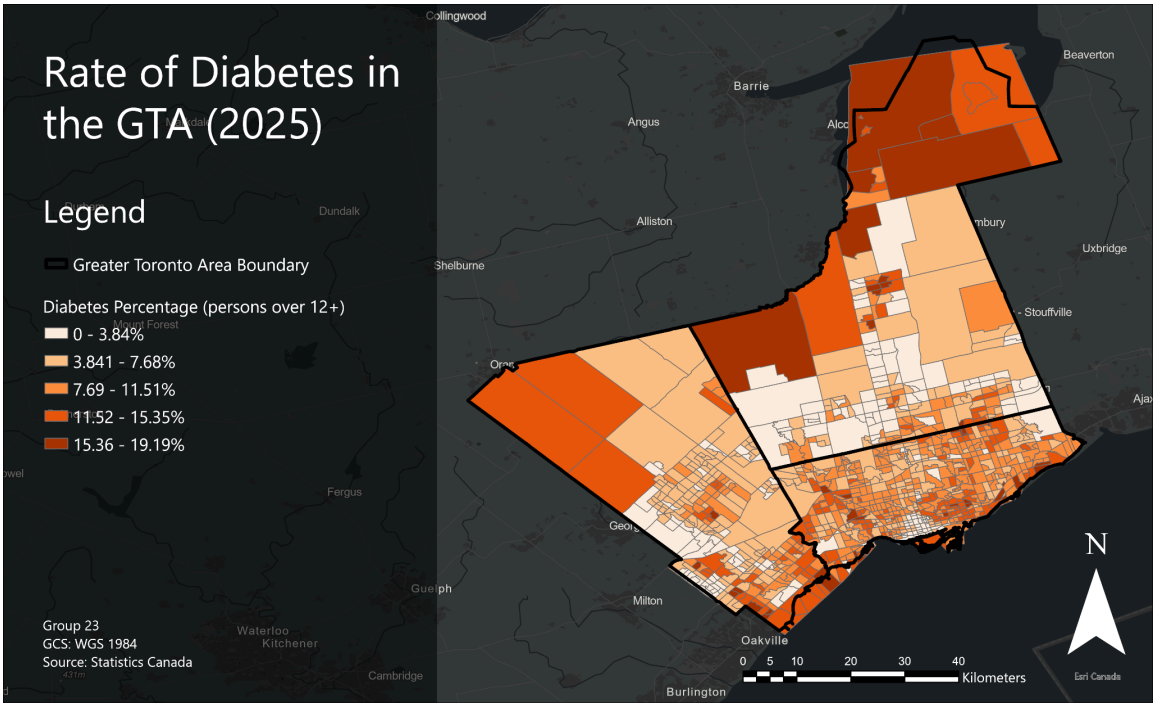


Figure 2 (above): Percentage of diabetes for persons above 12 years old in the Peel, York, and Toronto Regional Municipalities by census tract.

Figure 3 illustrates the distribution of diabetes prevalence across the Greater Toronto Area census tracts. The values spread from 0.0% to 19.1%, indicating that while some areas show

comparatively lower diabetes rates, there are other areas which tend to have very high prevalence. The median is about 7.8%, and interquartile range goes from approximately 4.4% to 10.7%, meaning that the middle 50% (25th percentile to 75th percentile) of the census tracts have diabetes rates falling between this range. The longer upper whisker highlights the presence of census tracts with a higher diabetes rate compared to others. The mean is 7.7%, which is similar to the median.

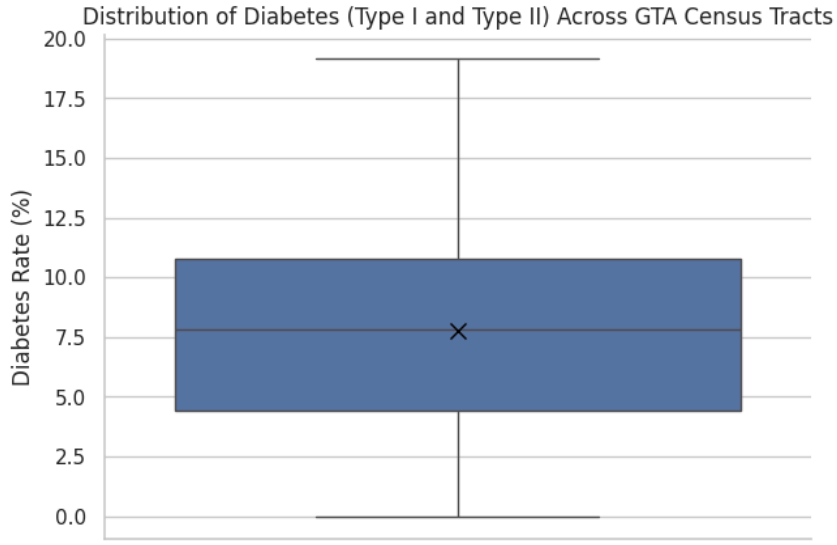


Figure 3 (above): Box plot of the diabetes prevalence by census tract for Peel, York, and Toronto Regional Municipalities in total.

Table 1 summarizes the diabetes rate across Peel, Toronto and the York regions. Toronto has the highest value for the mean diabetes percentage of 8.6%, with Peel and York having 6.8% and 6.4%, respectively. Similarly, Toronto has the highest median percentage value of 9.1% compared to 5.3% and 5.5% of Peel and York regions. These high values of mean and median explain that Toronto census tracts tend to have a higher prevalence of diabetes as compared to the other regions. On the other hand, Peel has the highest maximum diabetes percentage value at 19.1%, with York having 18.0%, and Toronto having the lowest at 17.9%. While Toronto has the highest average and the largest number of census tracts of around 585, compared to 282 in the Peel region and 198 in York, the values of all the regions tend to be very similar and display higher values.

Region	No. of Tracts	Mean Diabetes (%)	Max Diabetes (%)
Peel	282	6.80	19.19
Toronto	585	8.68	17.91
York	198	6.44	18.05

Table 1: Grouped summary for Toronto, York, and Peel Regions.

Figure 4 below displays the spatial distribution of each variable alone to provide a picture of the tracts before applying further analysis. The spatial autocorrelation each variable does suggest a degree of clustering and some trends that may or may not be correlated, e.g. higher diabetes percentages tend to appear in tracts that also have lower educational attainment, especially where the share with no certificate is higher and postsecondary education is lower. However, visual univariate inspection is not sufficient for drawing conclusions about spatial patterns.



Figure 4: Univariate choropleth maps of each variable.

Figures 5-7 show non-spatial correlation for diabetes against various levels of educational attainment and their respective scatter plots. There is a slight positive relationship between those who have earned no certifications and those who have earned only a high school diploma with diabetes, with the values being 0.3048412348957757 and 0.25670645608635884, respectively. On the other hand, there is a negative relationship between post-secondary education attainment and diabetes percentage per census tract, with a value of -0.3032641838771275.

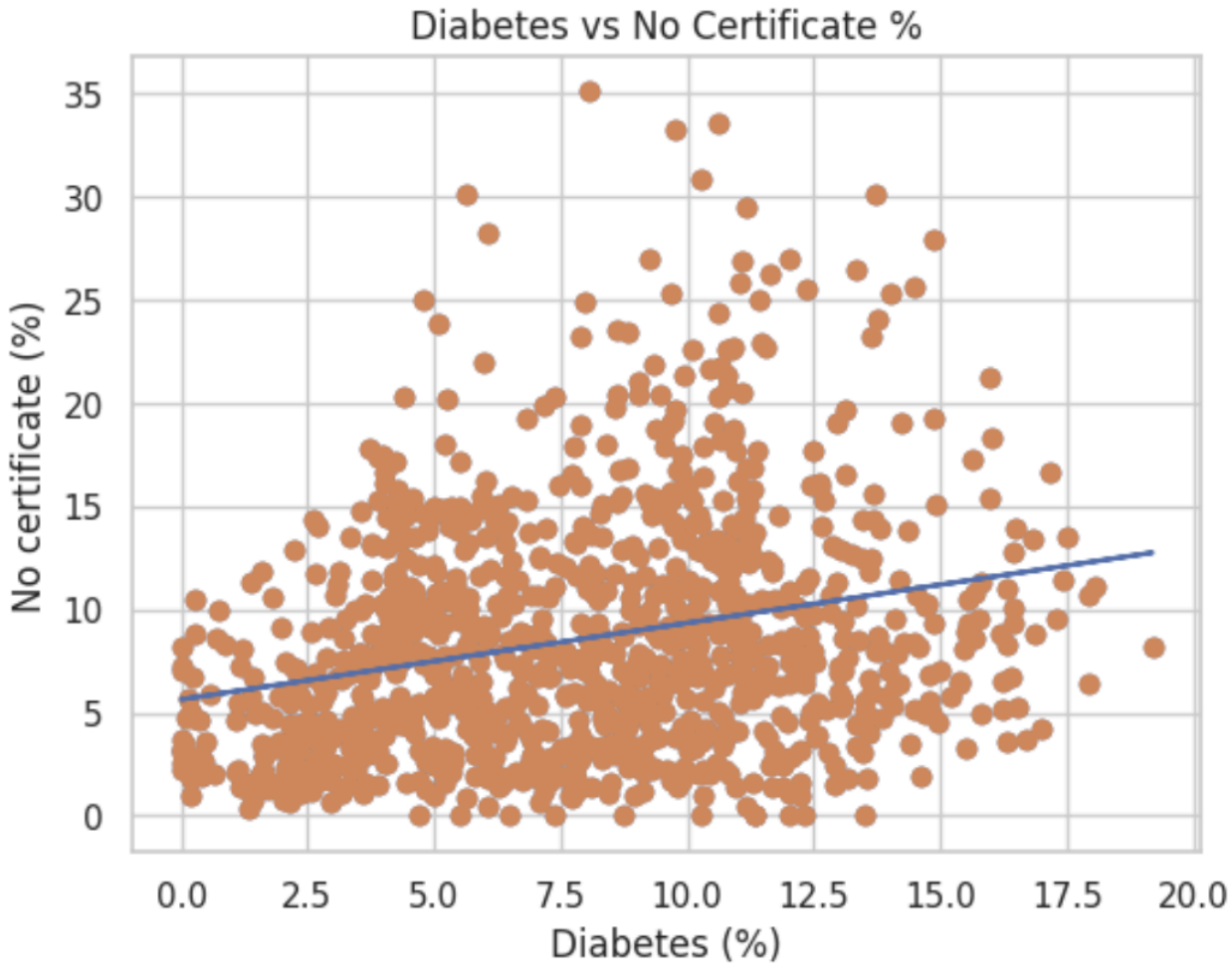


Figure 5: Non-spatial correlation chart for diabetes percentage and no certificate percentage.

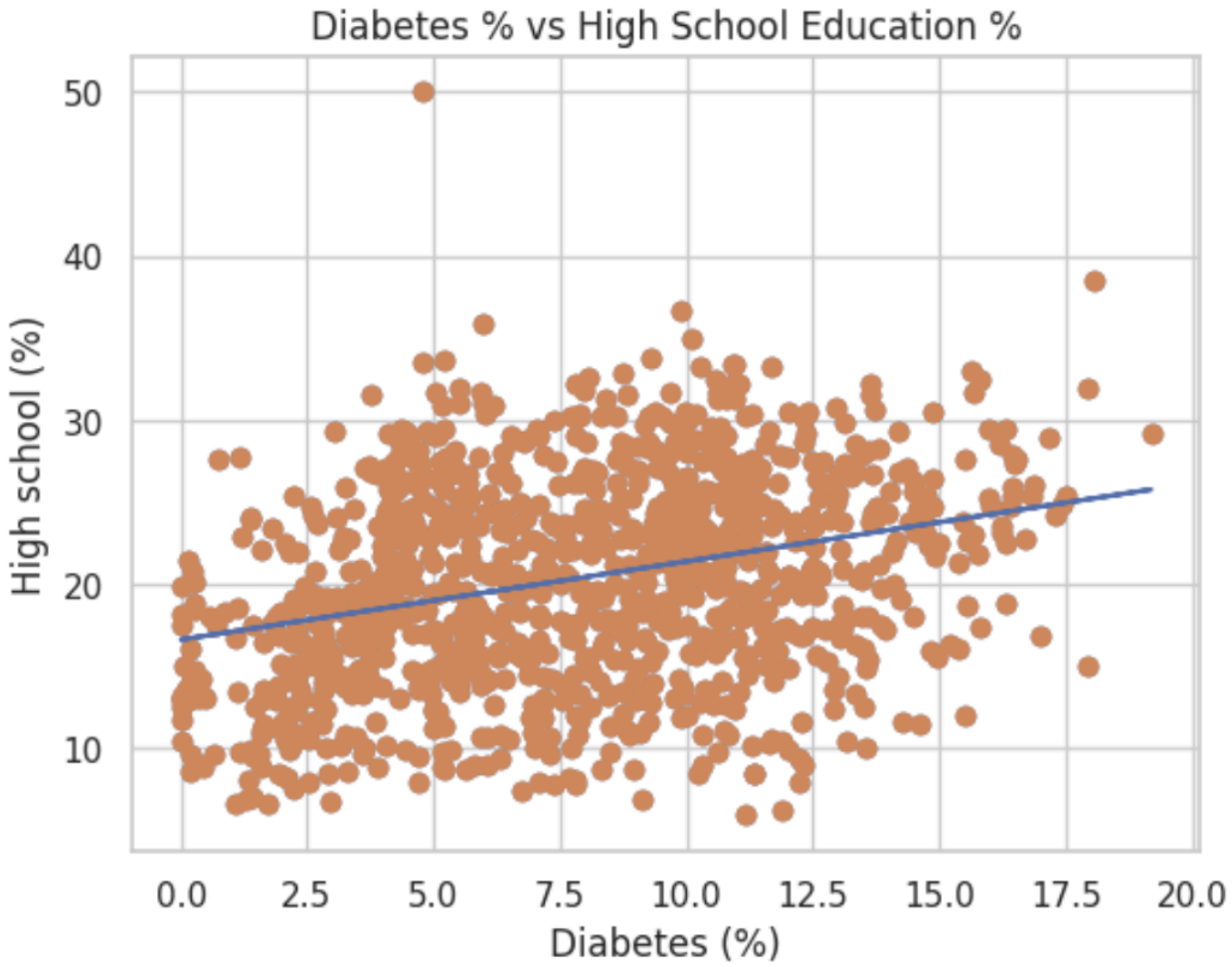


Figure 6: non-spatial correlation chart for diabetes percentage and high school education percentage.

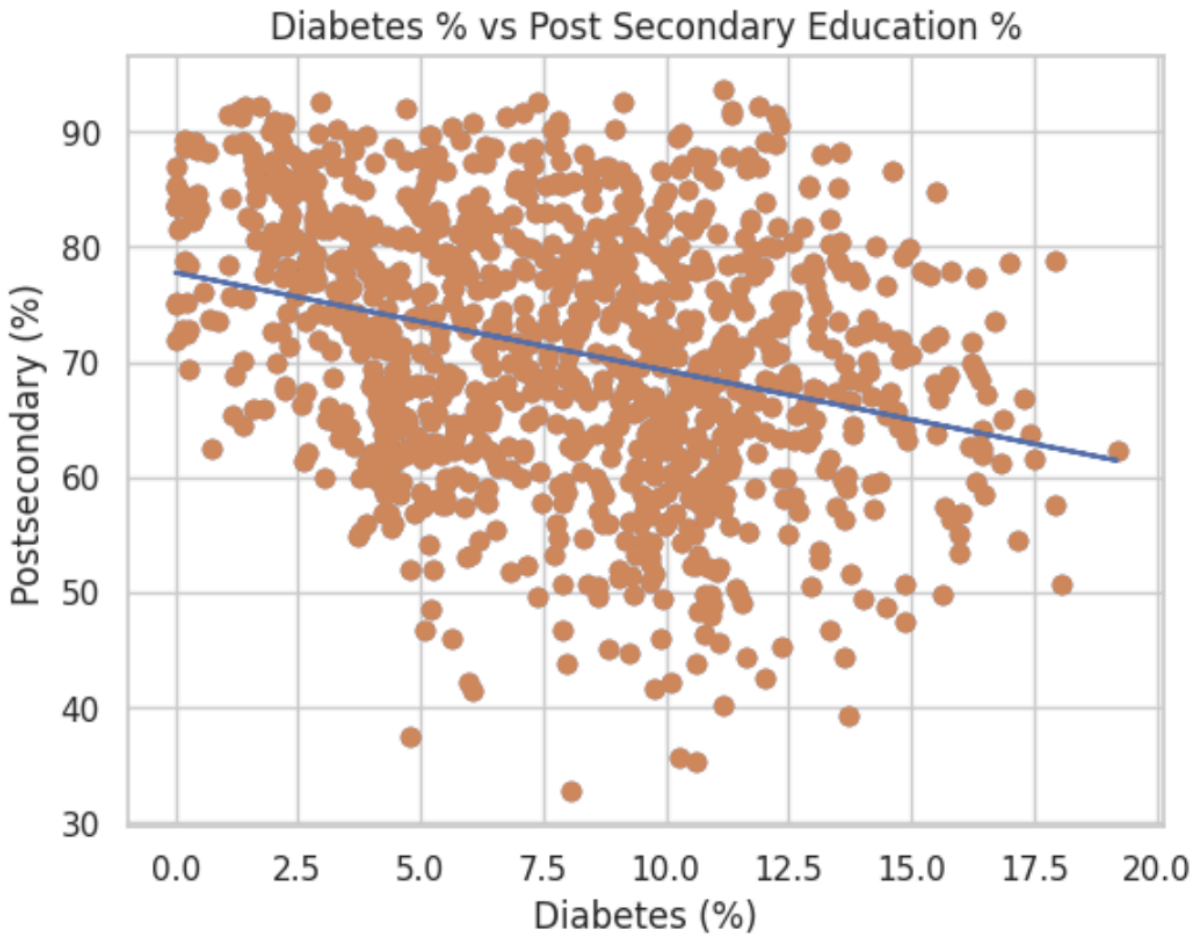


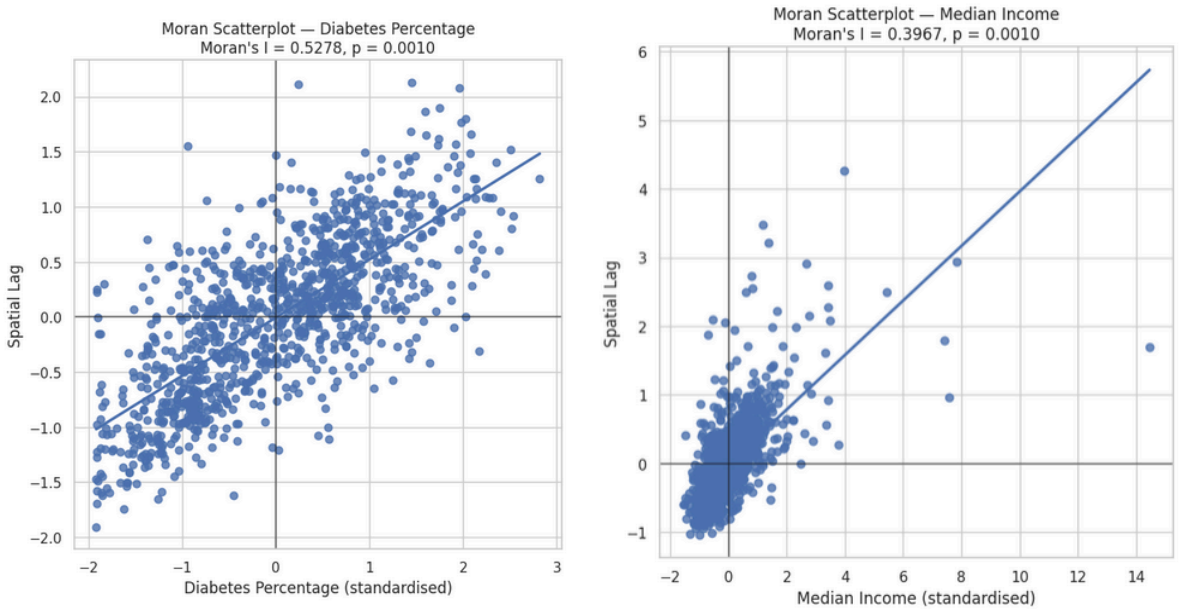
Figure 7: non-spatial correlation chart for diabetes percentage and post-secondary education percentage.

These differences across regions point to an uneven spatial distribution of diabetes, which motivated the spatial autocorrelation analysis that follows. All five variables showed statistically significant positive spatial autocorrelation (p = 0.001). The education variables clustered most strongly, with postsecondary education returning the highest Moran's I, followed by no certificate % and high school %. Diabetes and median income showed moderate but still significant clustering.

Variable	**Moran's I**	**z-score**	**p-value**	**Significant**
Diabetes Percentage	0.5278	31.05	0.001	Y

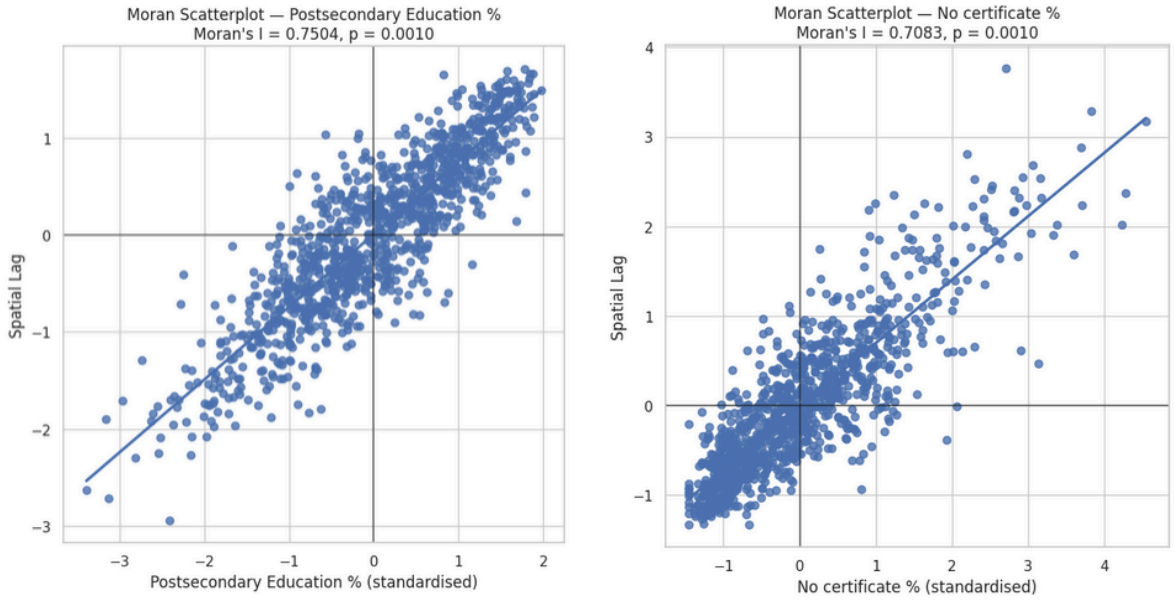
Median Income	0.3967	23.44	0.001	Y
Postsecondary Education %	0.7504	44.32	0.001	Y
No Certificate %	0.7083	41.01	0.001	Y
High (secondary) school %	0.7015	39.82	0.001	Y

Table 2: Global Moran's I Results.



a. Spatial Lag vs Diabetes Percentage **b. Spatial Lag vs Median Income**

Figure 6: Moran Scatterplot of Diabetes Percentage and Median Income.



a. Spatial Lag vs Postsecondary ***b. Spatial Lag vs No certificate***

Figure 7: Moran Scatterplot of Postsecondary Education % and No Certificate %.

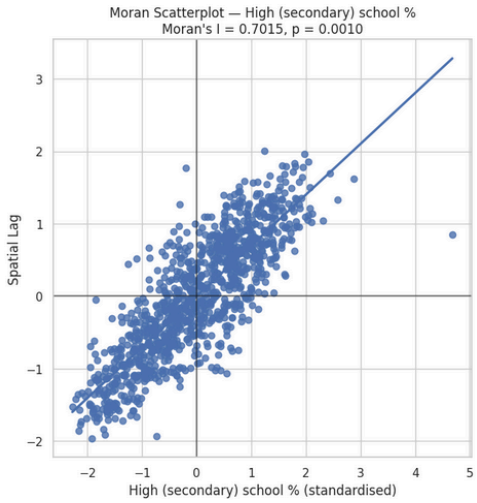
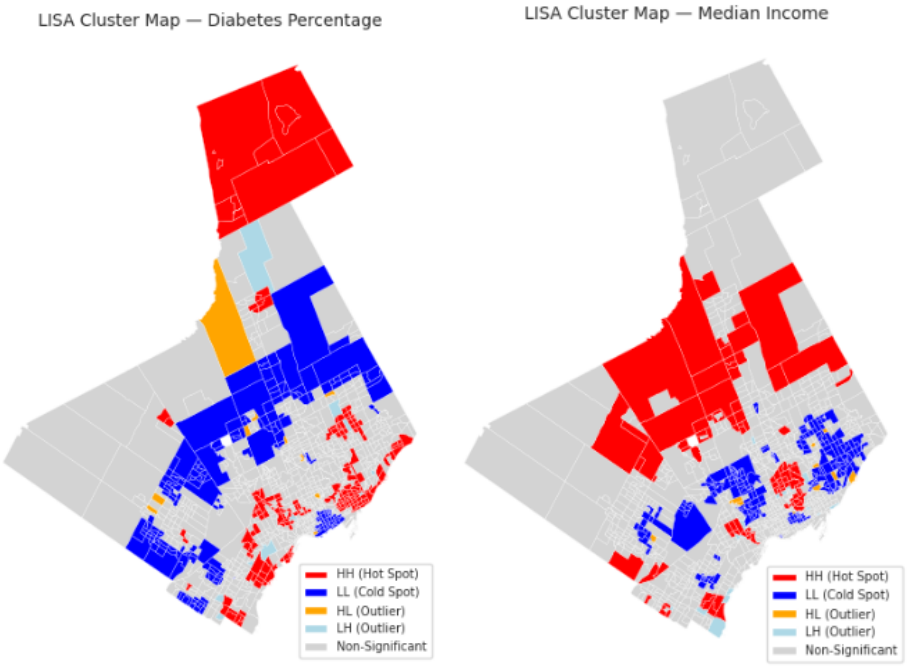


Figure 8: Moran Scatterplot of High (Secondary) School %

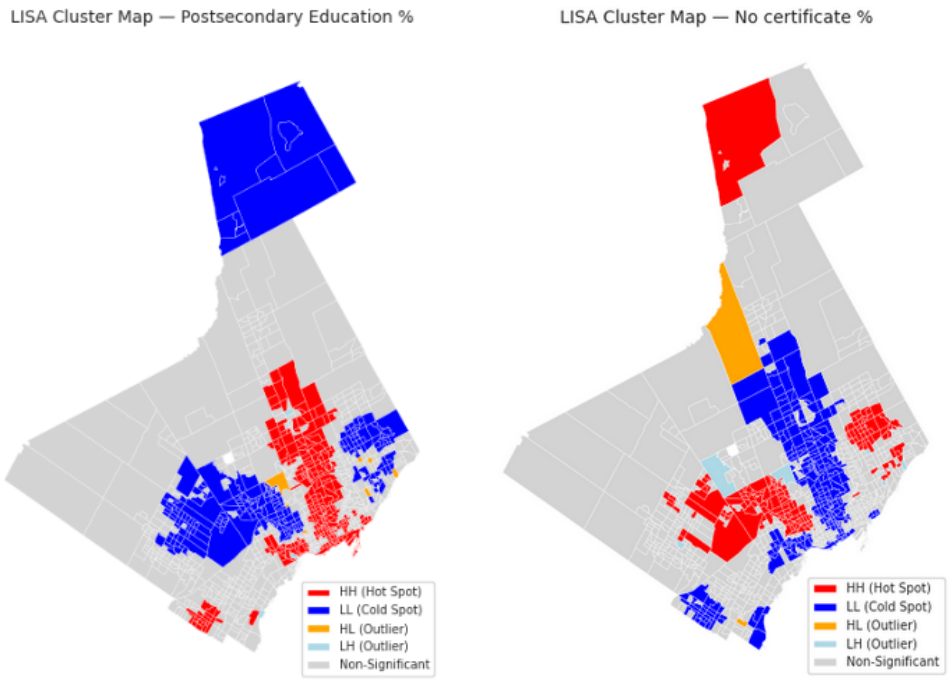
Looking at the LISA cluster maps (Figures X–X), a few clear geographic patterns stand out. Starting with diabetes, there is a large red HH hot spot sitting across the northern York region, such as King and East Gwillimbury, where high diabetes rates are consistently surrounded by more high diabetes tracts. Coming further south into Vaughan, the map gets more mixed, with an

orange HL outlier and blue LL cold spots appearing side by side, suggesting this part of York is more of a transitional zone. HH clusters then reappear in the eastern and inner city Toronto tracts in the bottom right of the map, while Peel and southern York are largely LL cold spots. The income map reads almost like the opposite of diabetes. HH clusters for income sit across Vaughan, Richmond Hill and Markham in York, while the LL cold spots land in Toronto and Brampton, the same areas where diabetes tends to be higher. The education maps reinforce this further. Postsecondary education has its strongest HH clustering across York, particularly Vaughan, Richmond Hill and Markham, with LL cold spots in Toronto and Brampton. No certificate % is almost the mirror image, with high clustering in Toronto and Brampton and low clustering in York. High school % shows strong HH clustering around the outer edges of the study area, particularly in northern York and parts of Peel, while LL cold spots sit more centrally. This is a different pattern from the other education variables and suggests that high school completion as the highest level of education tends to be more of a suburban and peripheral phenomenon across the GTA rather than concentrating in one specific region.



a. LISA Cluster Map - Diabetes Percentage **b. Spatial Lag vs Median Income**

Figure 9: LISA Cluster Map of Diabetes Percentage and Median Income.



a. LISA Cluster Map - Postsecondary *b. LISA Cluster Map - Certificate*

Figure 10: LISA Cluster Map of Postsecondary Education % and No Certificate %.

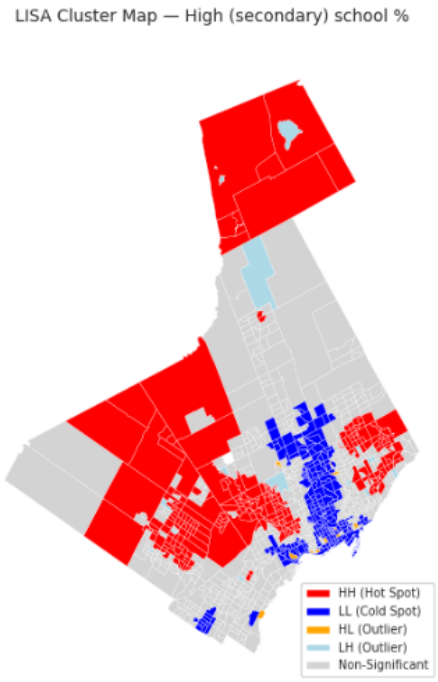


Figure 11: LISA Cluster Map of High (Secondary) School %

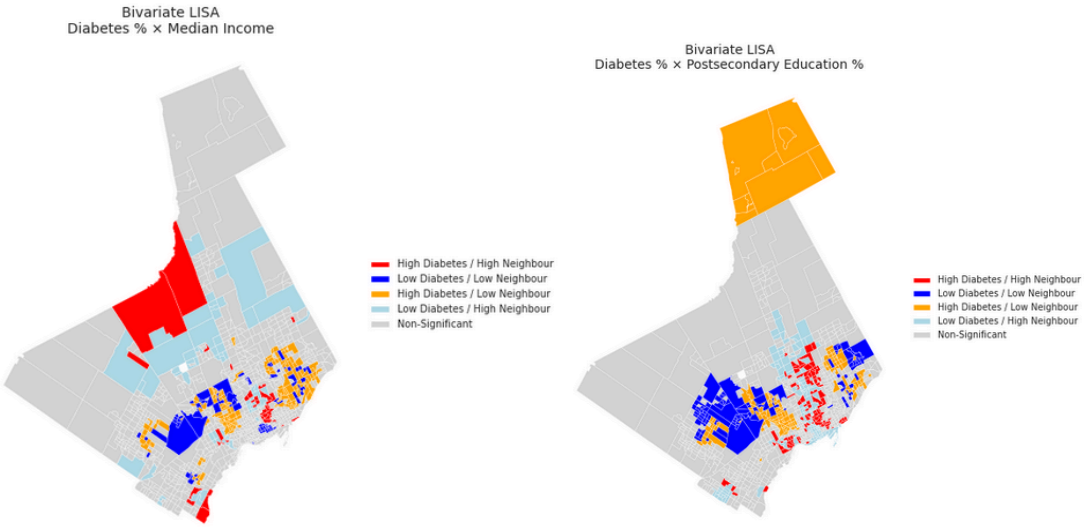
To build on the univariate LISA results, bivariate Moran's I was run to test whether diabetes prevalence in one tract is spatially associated with the predictor variables in neighbouring tracts. All four bivariate relationships were statistically significant (p = 0.001). Diabetes shows a negative spatial association with both income and postsecondary education, meaning high diabetes tracts tend to be surrounded by low income/low education neighbours. The relationship flips for no certificate % and high school %, where high diabetes tracts cluster with high values of those variables in neighbouring tracts.

Variable Pair	**Moran's I**	**z-score**	**p-value**	**Significant**
Diabetes % × Median Income	-0.1166	-4.98	0.001	Y
Diabetes % × Postsecondary Education %	-0.2100	-9.01	0.001	Y

Diabetes % × No Certificate %	0.2012	8.52	0.001	Y
Diabetes % × High (secondary) school %	0.1950	8.72	0.001	Y

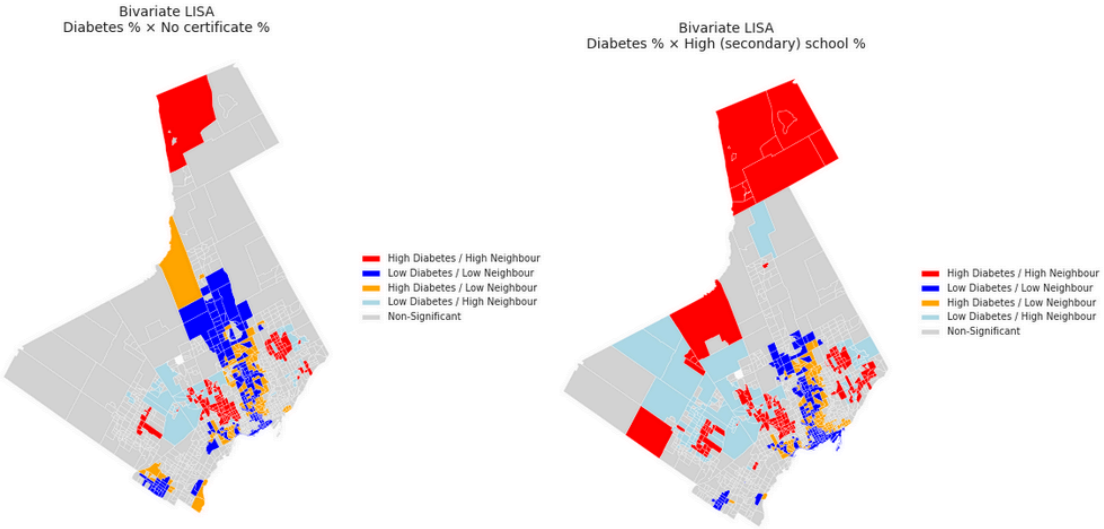
Table 3: Global Bivariate Moran's I Results

Looking at the bivariate LISA maps, the patterns are pretty consistent across all four predictors. For diabetes × median income (Figure X), central Toronto and Brampton are dominated by orange where high diabetes tracts are sitting next to low income neighbours while light blue Low Diabetes / High Neighbour tracts run along the Vaughan corridor, reflecting the sharp income divide between York and the inner city. The postsecondary map is the most striking. A large orange block swallows almost all of northern York, which makes geographic sense where King and East Gwillimbury are right next to the highly educated Vaughan and Richmond Hill tracts, so high diabetes areas end up bordered by high education neighbours. Central Toronto and Brampton meanwhile show a mix of red HH and blue LL. The no certificate % and high school % maps look similar to each other, with red HH clusters in northern York, Brampton and parts of Toronto. The high school map has the highest proportion of significant tracts at nearly 50%, and a notable stretch of light blue across western Peel where diabetes is low despite neighbouring tracts having high high school completion rates.



a. Bivariate LISA Diabetes vs Median Income ***b. Bivariate LISA Diabetes vs Postsecondary***

Figure 12: Bivariate LISA: Diabetes % × Median Income and Bivariate LISA: Diabetes % × Postsecondary Education %



a. Bivariate LISA Diabetes vs No certificate ***b. Bivariate LISA Diabetes vs Highschool***

Figure 13: Bivariate LISA: Diabetes % × No Certificate % and Bivariate LISA: Diabetes % × High (secondary) School %.

In Table 4, the regression results examine the relationship between different education levels and diabetes. The results show that no certification % and postsecondary % values are significant, however, high school % is no along with a low Rsquared value which is explained further in the discussion.

Variable	Coefficient	p-value	Significance	F-statistic
No certificate %	-0.57568	0.02887	Significant	
High school %	-0.44077	0.08640	Not significant	
Postsecondary %	-0.61471	0.01791	Significant	
R^2	0.099 (~10%)	N/A	Low	
Adjusted R^2				38.8440

Table 4: The table shows the results of the Ordinary Least Squares of the different education levels.

Table 5 presents the spatial lag model which is based on the concept of how nearby areas affect each other which makes them act similarly. Moreover, while no certification % and postsecondary % values are significant, however, high school % and diabetes are not.

Variable	Coefficient	p-value	Significance
No certificate	-0.59551	0.01689	Significant
High school	-0.45853	0.05909	Not significant
Postsecondary	-0.62491	0.01036	Significant
Spatial Lag (W_Diabetes)	0.12022	0.57843	Not significant
Pseudo R^2	0.2281	N/A	Moderate

Table 5: The table shows the Spatial Lag Model of the different education levels.

Through Table 5, it can be inferred that the spatial error model has been carried out which is associated with spatially correlated unobserved external factors. All the education show significant values,w ith lambda (λ) indicating a very strong spatial clustering that might be caused due to reasons other than education alone.

Variable	Coefficient	p-value	Significance
No certificate %	-0.93285	0.00000	Strong
High school %	-0.69994	0.00000	Strong
Postsecondary %	-0.94390	0.00000	Strong
Lambda (λ)	0.76748	0.00000	Strong spatial
Pseudo R^2	0.0985	N/A	Low

Table 6: The table shows the Spatial Error Model of the different education levels.

Figures 14-15 are the result of applying a k-means algorithm to cluster the data. Figure 14 shows the spatial distribution of the census tracts as ordinal clusters. Table 7 displays the cluster profiles showing the average value for diabetes percentage, post-secondary education percentage, etc. From Table 7, cluster 0 appears to be the most advantaged i.e. has the greatest median income, lowest percentage of diabetes, etc. Cluster 2 appears to be the most disadvantaged, least median income, etc.

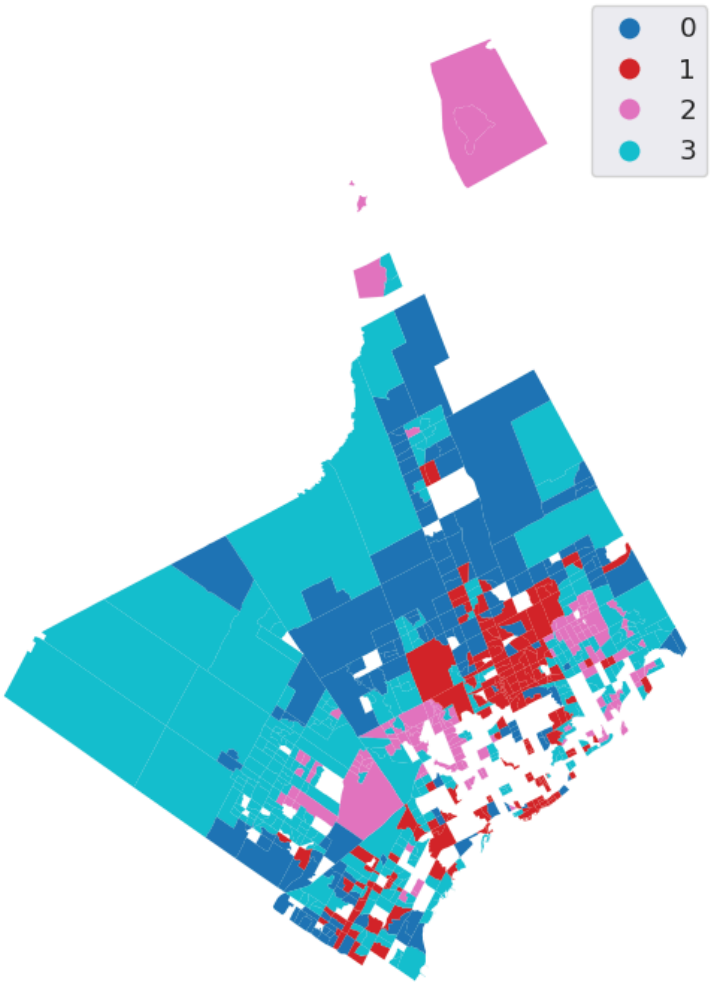


Figure 14: K-means cluster map (k=4).

Variable	0	1	2	3
Diabetes Percentage	3.786	6.514	9.566	8.765
Households by Income (Current Year) | Median Ho...	160639.282	99840.994	75312.427	99264.046

No certificate %	4.674	3.907	18.898	9.773
High (secondary) school %	16.924	14.537	28.178	23.887
Postsecondary Education %	78.382	81.528	52.987	66.335
Cluster sizes	153	188	128	311

Table 7: Cluster profile for k-means k= 4.



Figure 15: Cluster distribution for k-means (k=4).

Figures 16 and Table 8 are the findings of applying agglomerative hierarchical clustering as an alternative to k-means clustering. The clusters produced by Ward's hierarchical clustering share a similar spatial structure to those produced by k-means, but the cluster profiles reveal that the ordinal labels signify census tracts with different averages.

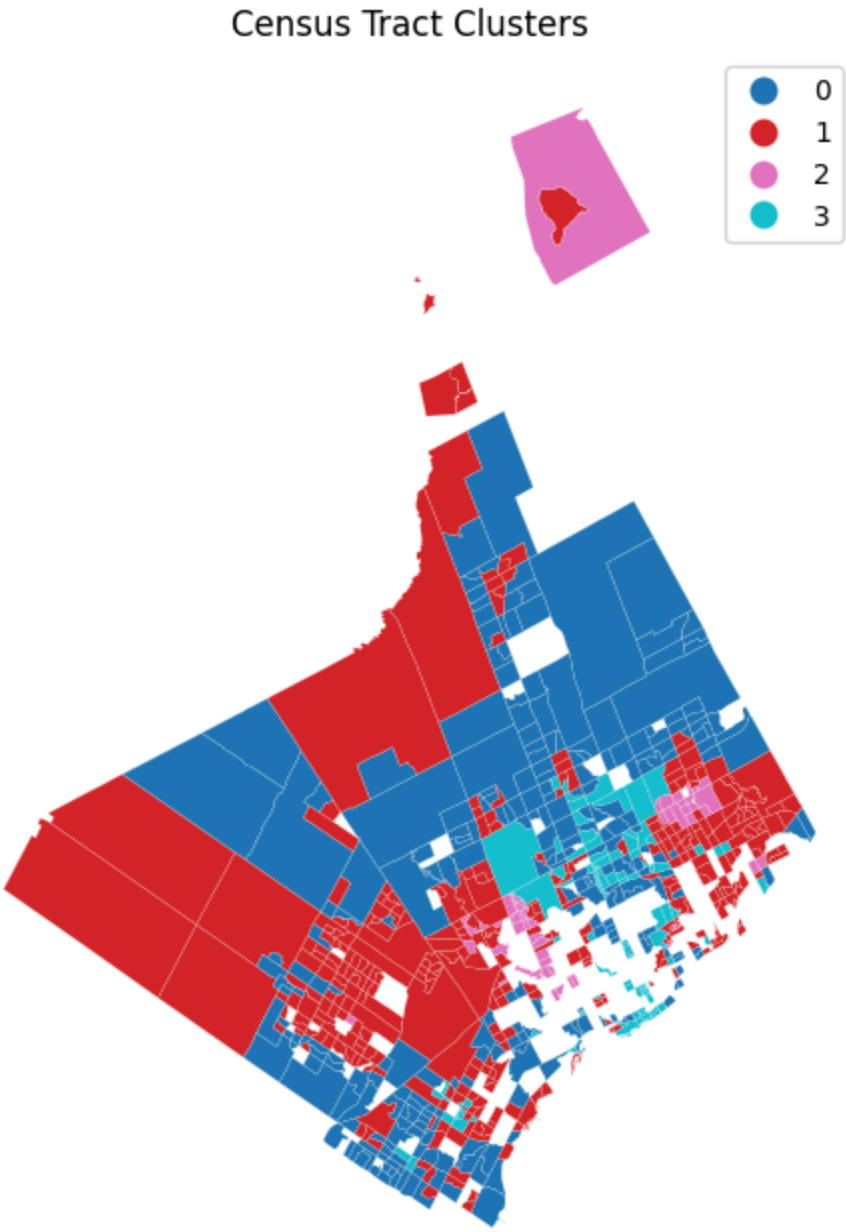


Figure 16: Cluster profile for Ward's hierarchical method n=4

Variable	0	1	2	3
Diabetes Percentage	3.786	6.514	9.566	8.765

Households by Income (Current Year) | Median Ho...	160639.282	99840.994	75312.427	99264.046
No certificate %	4.674	3.907	18.898	9.773
High (secondary) school %	16.924	14.537	28.178	23.887
Postsecondary Education %	78.382	81.528	52.987	66.335

Table 8: Cluster profiles for Ward's hierarchical method.

5 DISCUSSION & INTERPRETATION

The results illustrate the significant association of education with diabetes prevalence across the census tracts in the study area. Through Figures (5-7), it can be inferred that the scatterplots show a clear trend between different education levels and diabetes prevalence. Figure 5 shows a positive relationship between the two variables, as it shows that with the increase in the percentage of population with no educational certification, possibly due to the type of opportunities they got, and in case they dropped out of high school, there is an increase in the percentage of diabetes prevalence. This makes sense as this population may not have been provided with enough social awareness about diabetes, and steps to have a balanced diet and a healthy lifestyle to prevent getting the disease, as they did not get the opportunity to go to an academic institution and get information regarding it. Similarly, Figure 6, shows a positive trend, which shows how with the increase in the percentage of population of those completing high school education, there is an increase in the percentage of diabetes prevalence. Hence, further highlighting how just the completion of high school is not sufficient to be well-versed about the causes and prevention of diabetes (Luo et al., 2025). Figure 7 depicts the negative correlation between the percentage of the population who completed post-secondary education and the percentage of the population that has diabetes. This aligns with the idea that with the decrease in the percentage of the population completing postsecondary, the prevalence of diabetes across these regions tends to increase. This is further supported in the spatial analysis from figures 8-12 and tables 2-3. The results consistently point to a geographic divide, where high diabetes tracts in

northern York and inner city Toronto overlap with low income and low education zones. The bivariate analysis strengthens this where postsecondary education has the strongest spatial relationship with diabetes of all variables tested, while high school completion alone does not appear to offer the same protective effect. Hence, highlighting the importance of the completion of postsecondary education and the major role it plays when it comes to making people socially aware about diabetes, wellness and a healthy lifestyle. That said, the relatively weak Moran's I values for diabetes compared to the education variables indicate that while these relationships are spatially significant, diabetes distribution is likely shaped by a broader set of factors beyond education and income alone.

Figures 14-16 and tables 7-8 from the clustering algorithms reveal the spatial distribution of categories of clusters and their average values. The clusters are not contiguous and are disjointed, but the results are also not clear because of the numerous missing census tracts in the data, this limitation is addressed in the limitations section. Both clustering algorithms have produced similar overall spatial structure categorically and similar cluster profiles as well. However, the cluster distribution table and cluster profiles reveal that values are similar to each other and contain large amounts of overlap between ranges. This may indicate that because clustering algorithms are not producing starkly different cluster profiles for both algorithms, clustering may not be the best approach and the results may not be very informative.

The null hypothesis: Education does not have an effect on the spatial distribution of diabetes. Table 3 shows that the p-value of the no certification percentage population is around 0.028, which is less than 0.05, hence indicating that there is only 2.8% chance for the results to occur randomly, which basically implies that the relationship between the variables is significant. This means that areas which have a higher percentage of no certification groups will tend to have more diabetes, hence indicating a positive correlation between the two variables. Similarly, post-secondary percentage population has a p-value of 0.017, which is less than 0.05. This shows that there is only 1.7% chance for these results to occur randomly, hence indicating a statistically strong relationship between the variables. This means that with the decrease in the percentage of population who have completed postsecondary education, there is an increase in the diabetes prevalence which is illustrated by the negative correlation. Conversely, the high school percentage of people has a p value of around 0.086, which is more than 0.05, hence further indicating that the variables show a weak positive relationship, as there is 86% of chance for the

results to occur randomly. All these results tell that the no certificate percentage and postsecondary education percentage reject the null hypothesis, whereas the analysis fails to reject the null hypothesis for high school percentage, suggesting a weaker or insignificant relationship int hat specific education level. The R-squared value is 0.099, meaning that education overall explains only 10% of the variation of diabetes prevalence percentage across the census tracts and indicates that there are other spatial observational factors which affect diabetes distribution more. However, the variable should still be considered as we have 1059 census tracts, which is a huge sample size, so even a small weak relationship will be statistically significant as indicated by the f-statistics as well, which is 38.844. Hence, implying that the variables have a weak but statistically significant relationship. Table 4 showcases the spatial lag model, which works on the principle of nearby regions acting similarly, meaning that the diabetes rates in one tract will have a direct influence on the neighbouring tracts as well. The model shows significant values for no certificate and postsecondary percentages as their p values are less than 0.05 i.e., 0.01689 and 0.01036 respectively, them significant but not significant for highschool and diabetes due to higher p value. Conversely, Table 5 illustrates the spatial error model, which implies that the neighbouring tracts tend to be similar not because of the closeness based on distance between the tracts but due to various other unmeasured background conditions that cluster geographically. In this case, the spatial error model is more accurate as diabetes does not work in terms of the proximity of neighbourhoods to each other, but is 90% impacted by other factors like environmental or socioeconomic. Furthermore, the lambda coefficient ($\lambda = 0.767$) is large and highly significant, which confirms that the nearby regions are connected with each other due to other reasons and not spatial connectivity. This implies that for this analysis, the spatial error model provides a more theoretically appropriate and statistically justified choice for understanding the relationship between the variables across the GTA census tracts.

6 LIMITATION & UNCERTAINTY

When interpreting the results of this study, several limitations should be considered. First, the datasets used were collected from different years due to availability constraints, including diabetes and income data from 2025, recreational facilities data from 2021, and education data from 2021 census sources. This temporal mismatch may reduce the accuracy of the relationships observed, as these variables may not fully align across time.

In addition, missing values across datasets resulted in the exclusion of some census tracts during clustering analysis. This reduced the overall spatial coverage of the study area and may have influenced the observed patterns and cluster distributions. Another important limitation is that as the analysis is conducted at the census tract level, it introduces the risk of ecological fallacy meaning that the relationships identified reflect area-level trends and cannot be directly applied to individual-level behaviour or health outcomes.

Furthermore, while education was found to be statistically significant, the overall explanatory power of the models remained low, as indicated by the low R^2 values. This result can be interpreted as an example of ecological fallacy, since the relationship observed at the aggregated level does not necessarily reflect individual-level patterns. This suggests that education captures only a portion of the variation in diabetes prevalence, and that other unmeasured factors such as diet, ethnicity, healthcare access, and lifestyle likely play a substantial role. To further explore these influences, additional variables such as median income and recreational facility accessibility were tested. However, these variables did not produce strong or significant model results, reinforcing the complexity of diabetes as a health outcome and highlighting that it cannot be explained by a single or limited set of socioeconomic factors.

7 CONCLUSION

To conclude, it can be inferred that overall, the null hypothesis is rejected, as the results indicate that education is significantly associated with diabetes prevalence across census tracts in Toronto, Peel, and York regions. Areas with lower educational attainment, particularly higher proportions of populations with no certification, tend to exhibit higher levels of diabetes, whereas higher levels of postsecondary education are associated with lower diabetes prevalence.

The analysis further suggests that diabetes prevalence exhibits statistically significant spatial clustering within the study area, indicating that these patterns are not randomly distributed but are influenced by underlying social and environmental conditions. We tested both spatial lag and spatial error models, but selected the spatial error model as more appropriate, as similarities between neighbouring areas are better explained by shared underlying factors rather than direct spatial influence.

However, despite the statistical significance of education, the relatively low explanatory power of the models indicates that education alone explains only a small portion of the variation in diabetes prevalence. Additional variables such as income and recreational accessibility were tested but did not produce strong or significant results, highlighting the complexity of diabetes as a multifactorial health outcome. Overall, this study demonstrates the importance of education in understanding spatial health inequalities, while also emphasizing the need to consider a broader range of factors to more fully explain the distribution of diabetes across the study area.

8 REFERENCES

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Appendix

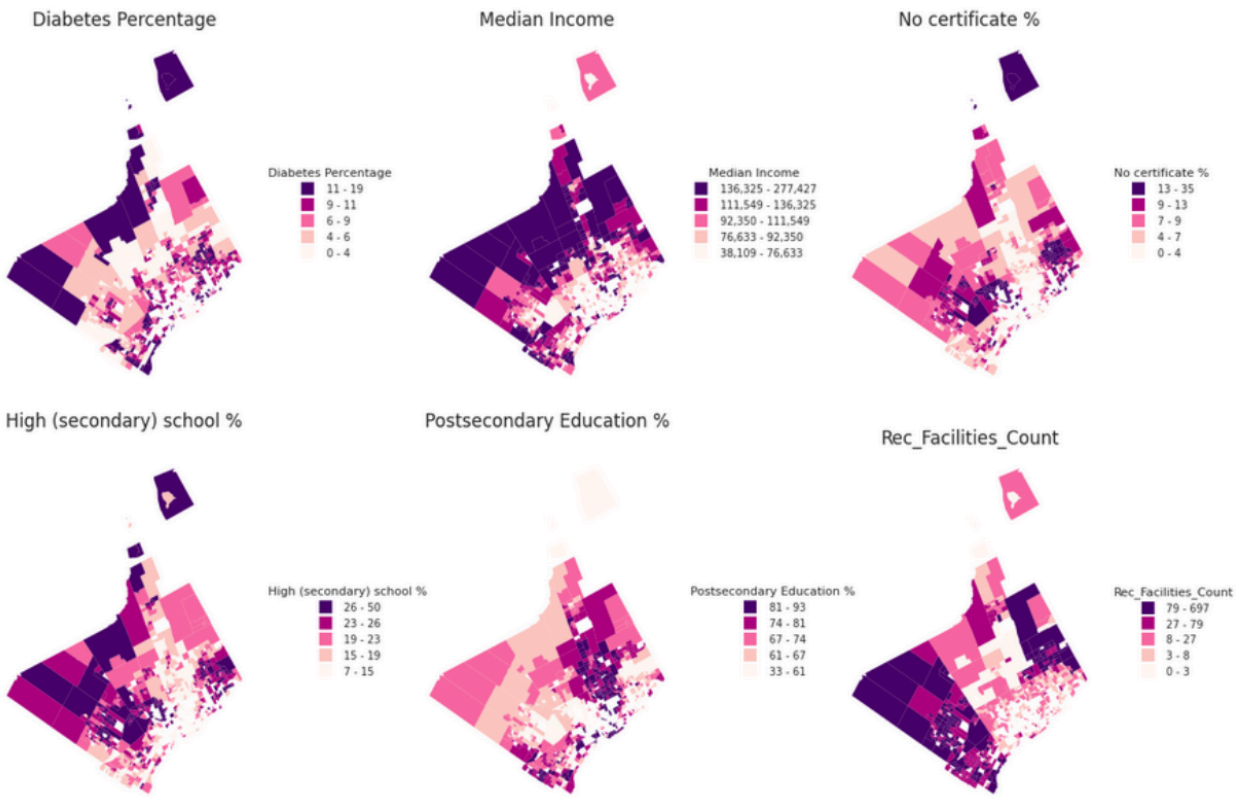


Figure 1: Spatial distribution of each variable

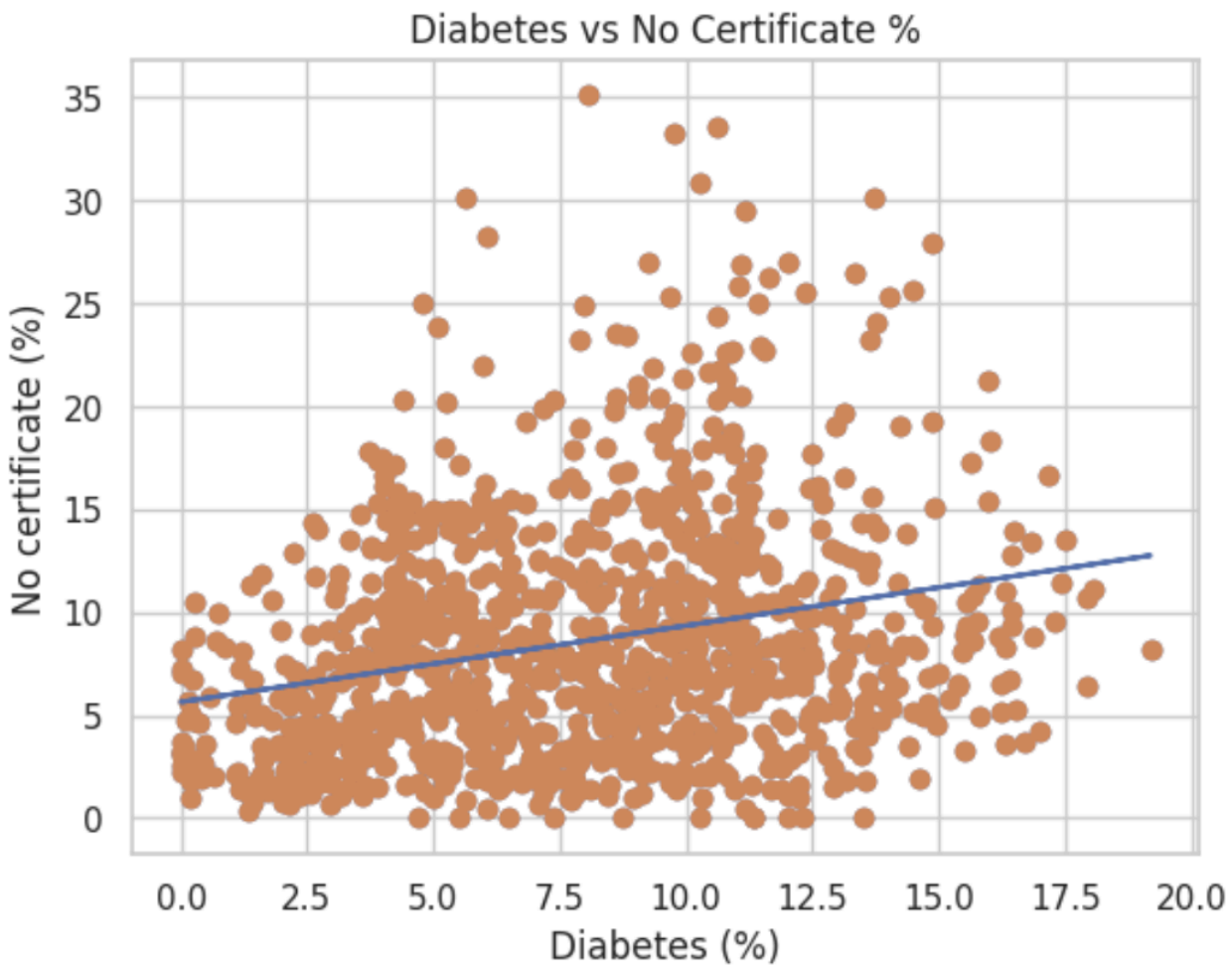

Diabetes vs No Certificate %
No certificate (%)
Diabetes (%)
0
5
10
15
20
25
30
35
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0

Figure 5:

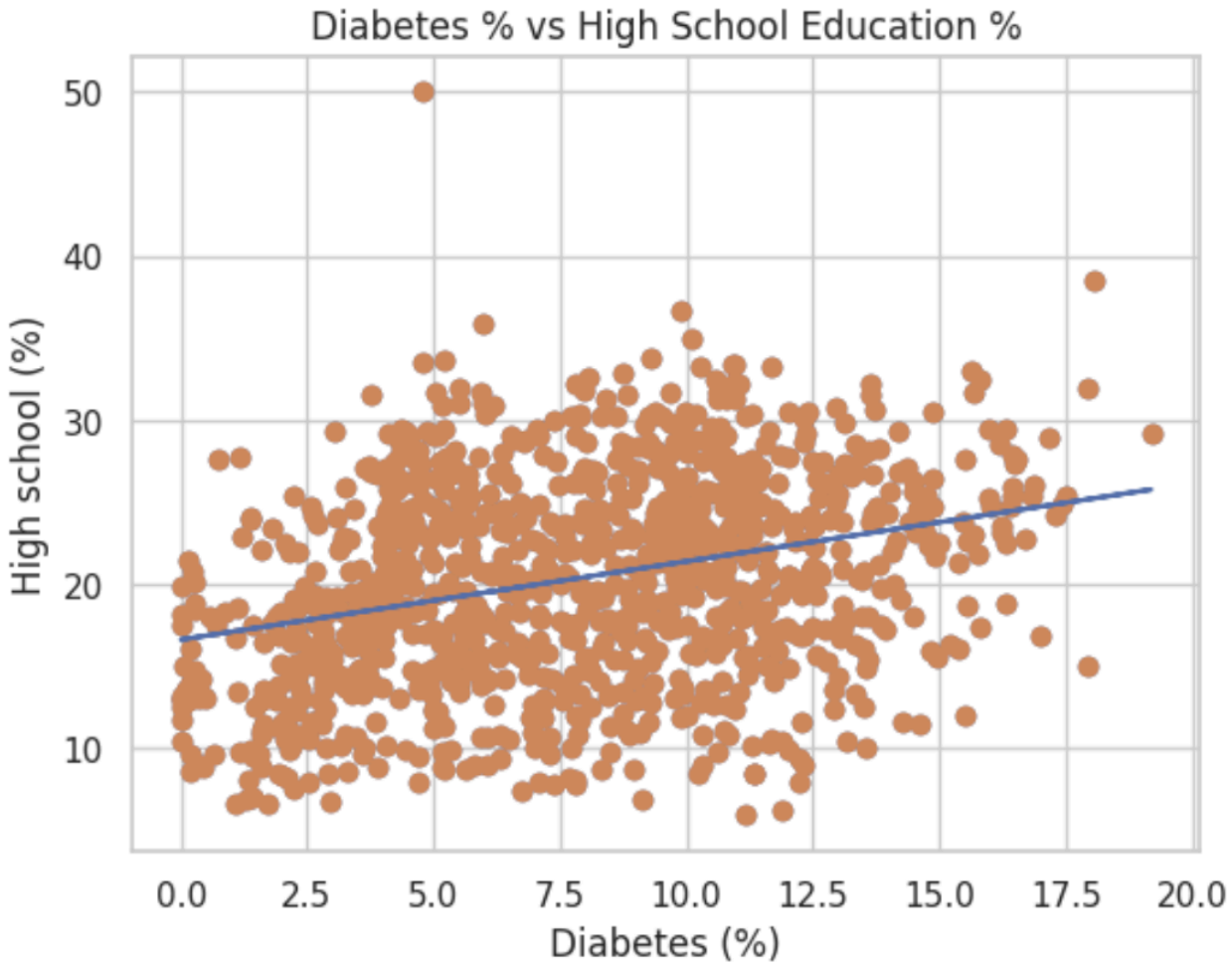

Diabetes % vs High School Education %
High school (%)
50
40
30
20
10
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Diabetes (%)

Figure 6:

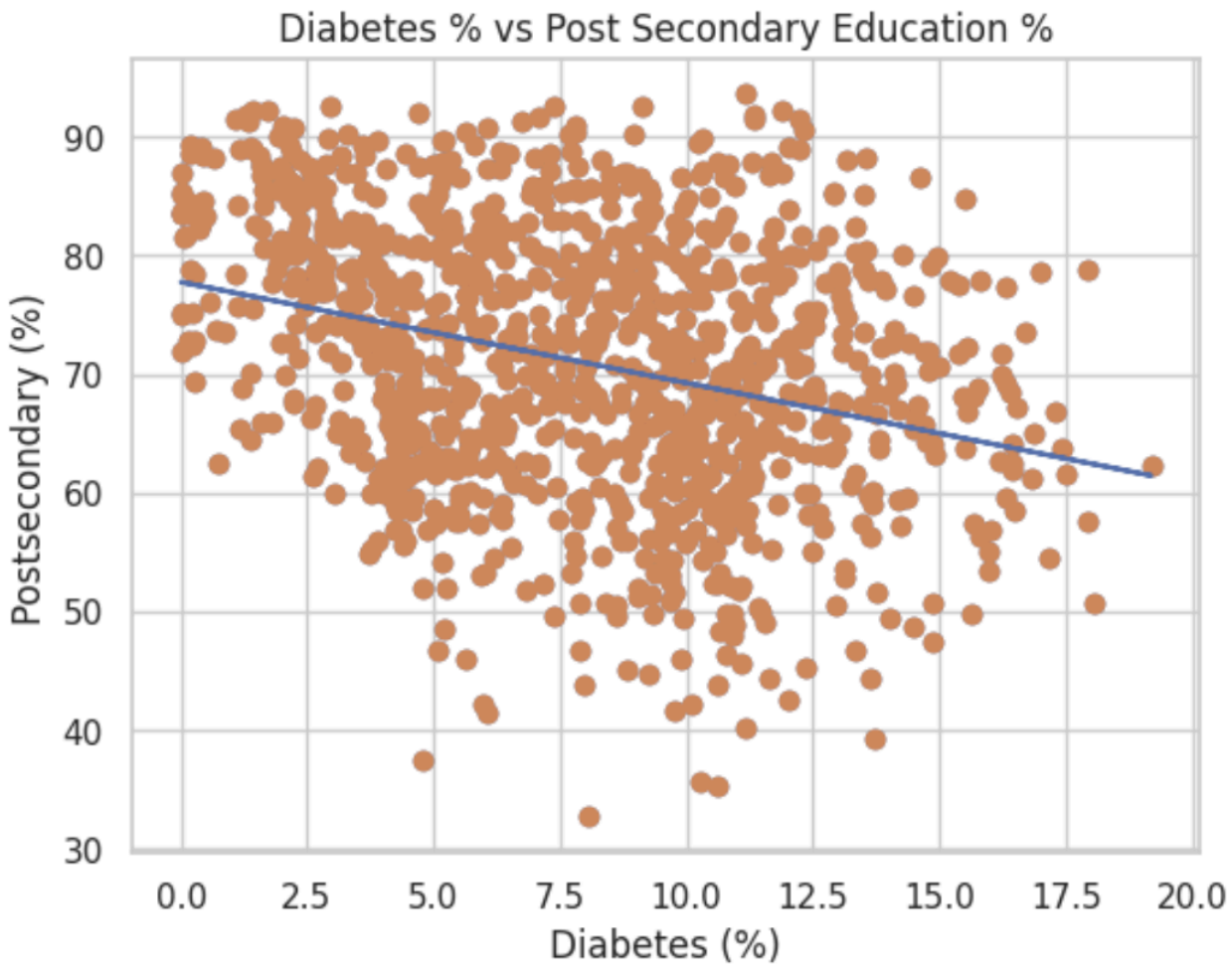

Diabetes % vs Post Secondary Education %
Postsecondary (%)
90
80
70
60
50
40
30
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Diabetes (%)

Figure 7:

```
REGRESSION RESULTS
------------------

SUMMARY OF OUTPUT: ORDINARY LEAST SQUARES
-----------------------------------------------------------------------------
Data set            :     unknown
Weights matrix      :        None
Dependent Variable  :    Diabetes                Number of Observations:        1059
Mean dependent var  :      7.7712                Number of Variables   :           4
S.D. dependent var  :      4.0511                Degrees of Freedom    :        1055
R-squared           :      0.0995
Adjusted R-squared  :      0.0969
Sum squared residual:     15635.9                F-statistic           :     38.8440
Sigma-square        :      14.821                Prob(F-statistic)     :   8.208e-24
S.E. of regression  :       3.850                Log likelihood        :   -2928.198
Sigma-square ML     :      14.765                Akaike info criterion :    5864.396
S.E of regression ML:      3.8425                Schwarz criterion     :    5884.256

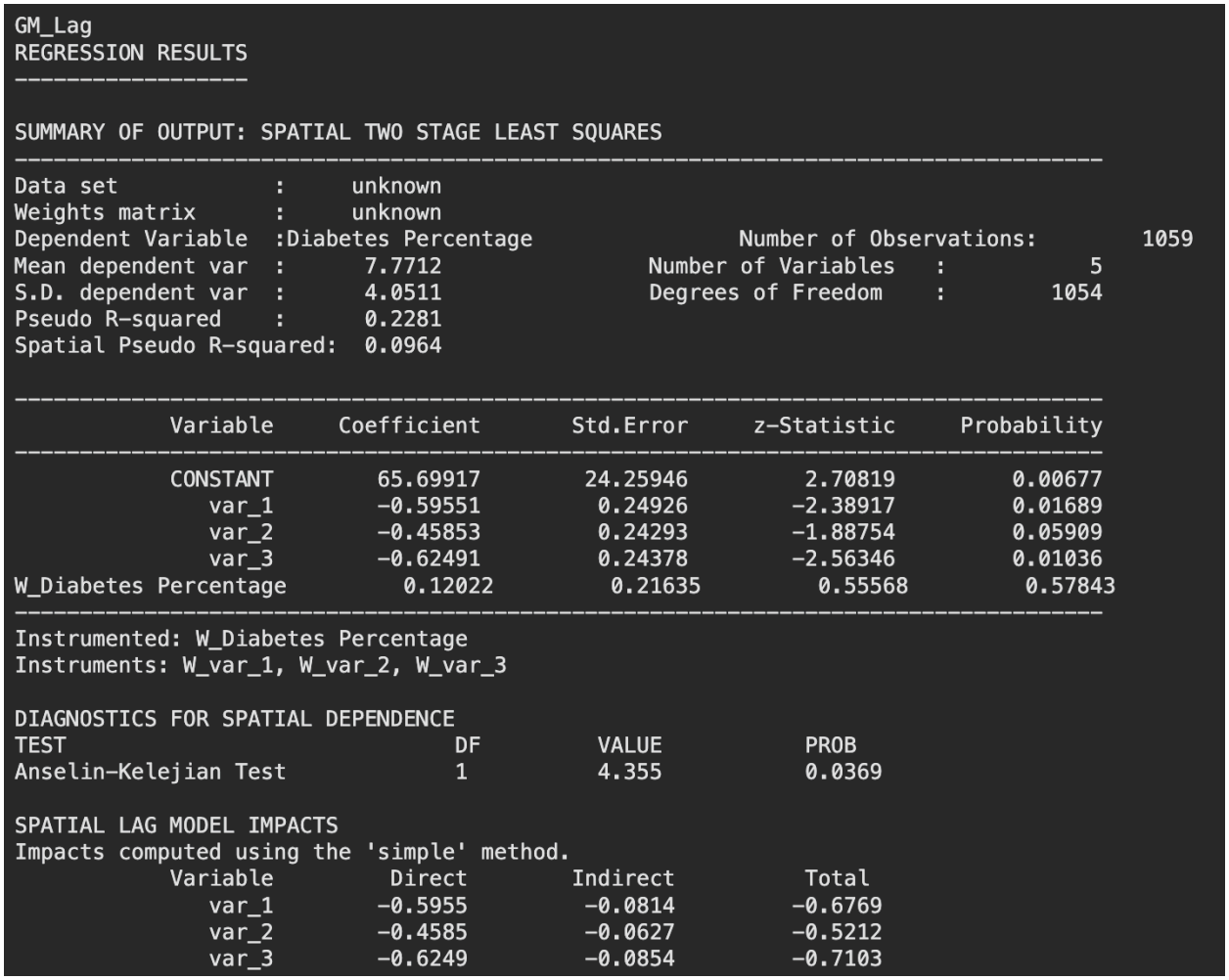
-----------------------------------------------------------------------------
            Variable     Coefficient       Std.Error     t-Statistic     Probability
-----------------------------------------------------------------------------
            CONSTANT        65.37418        25.86367         2.52765         0.01163
    No certificate %        -0.57568         0.26308        -2.18827         0.02887
High (secondary) school %        -0.44077         0.25681        -1.71630         0.08640
Postsecondary Education %        -0.61471         0.25923        -2.37125         0.01791
-----------------------------------------------------------------------------

REGRESSION DIAGNOSTICS
MULTICOLLINEARITY CONDITION NUMBER         525.162

TEST ON NORMALITY OF ERRORS
TEST                             DF        VALUE           PROB
Jarque-Bera                       2         35.560           0.0000

DIAGNOSTICS FOR HETEROSKEDASTICITY
RANDOM COEFFICIENTS
TEST                             DF        VALUE           PROB
Breusch-Pagan test                3         11.874           0.0078
Koenker-Bassett test              3         18.634           0.0003
```

Figure 8:



```
GM_Lag
REGRESSION RESULTS
------------------

SUMMARY OF OUTPUT: SPATIAL TWO STAGE LEAST SQUARES
--------------------------------------------------------------------------------
Data set            :     unknown
Weights matrix      :     unknown
Dependent Variable  :Diabetes Percentage               Number of Observations:        1059
Mean dependent var  :      7.7712               Number of Variables   :           5
S.D. dependent var  :      4.0511               Degrees of Freedom    :        1054
Pseudo R-squared    :      0.2281
Spatial Pseudo R-squared:  0.0964

------------------------------------------------------------------------------------
            Variable     Coefficient       Std.Error     z-Statistic     Probability
------------------------------------------------------------------------------------
            CONSTANT        65.69917        24.25946         2.70819         0.00677
               var_1        -0.59551         0.24926        -2.38917         0.01689
               var_2        -0.45853         0.24293        -1.88754         0.05909
               var_3        -0.62491         0.24378        -2.56346         0.01036
W_Diabetes Percentage         0.12022         0.21635         0.55568         0.57843
------------------------------------------------------------------------------------
Instrumented: W_Diabetes Percentage
Instruments: W_var_1, W_var_2, W_var_3

DIAGNOSTICS FOR SPATIAL DEPENDENCE
TEST                           DF         VALUE           PROB
Anselin-Kelejian Test          1          4.355           0.0369

SPATIAL LAG MODEL IMPACTS
Impacts computed using the 'simple' method.
            Variable         Direct        Indirect          Total
               var_1        -0.5955         -0.0814         -0.6769
               var_2        -0.4585         -0.0627         -0.5212
               var_3        -0.6249         -0.0854         -0.7103
```

Figure 6:

```
GM_Error_Het
REGRESSION RESULTS
------------------

SUMMARY OF OUTPUT: GM SPATIALLY WEIGHTED LEAST SQUARES (HET)
-----------------------------------------------------------------------------
Data set            :     unknown
Weights matrix      :     unknown
Dependent Variable  :Diabetes Percentage              Number of Observations:        1059
Mean dependent var  :      7.7712               Number of Variables   :           4
S.D. dependent var  :      4.0511               Degrees of Freedom    :        1055
Pseudo R-squared    :      0.0985
N. of iterations    :           1               Step1c computed       :          No

------------------------------------------------------------------------------
            Variable     Coefficient       Std.Error     z-Statistic     Probability
------------------------------------------------------------------------------
            CONSTANT        97.13490        12.21606         7.95141         0.00000
    No certificate %        -0.93285         0.13482        -6.91917         0.00000
High (secondary) school %        -0.69994         0.11405        -6.13705         0.00000
Postsecondary Education %        -0.94390         0.12402        -7.61079         0.00000
              lambda         0.76748         0.02486        30.87375         0.00000
```

Figure 7:

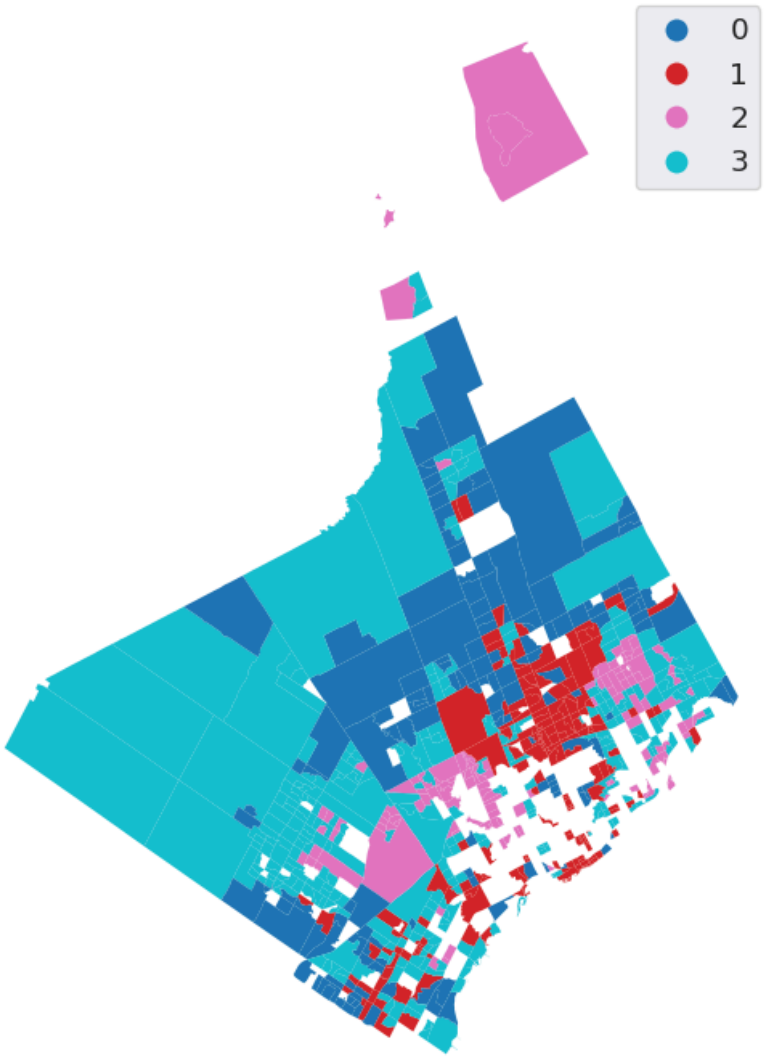


Figure 8: K-means cluster map (k=4)



Figure 9: Cluster distribution for k-means (k=4)

```
Cluster sizes:
k4cls
0    153
1    188
2    128
3    311
dtype: int64

Cluster means:
k4cls                                                    0          1
Diabetes Percentage                                  3.786      6.514
Households by Income (Current Year) | Median Ho...  160639.282  99840.994
No certificate %                                     4.674      3.907
High (secondary) school %                           16.924     14.537
Postsecondary Education %                           78.382     81.528

k4cls                                                    2          3
Diabetes Percentage                                  9.566      8.765
Households by Income (Current Year) | Median Ho...  75312.427  99264.046
No certificate %                                    18.898      9.773
High (secondary) school %                           28.178     23.887
Postsecondary Education %                           52.987     66.335
```

Figure 10: Cluster profile for k-means = 4

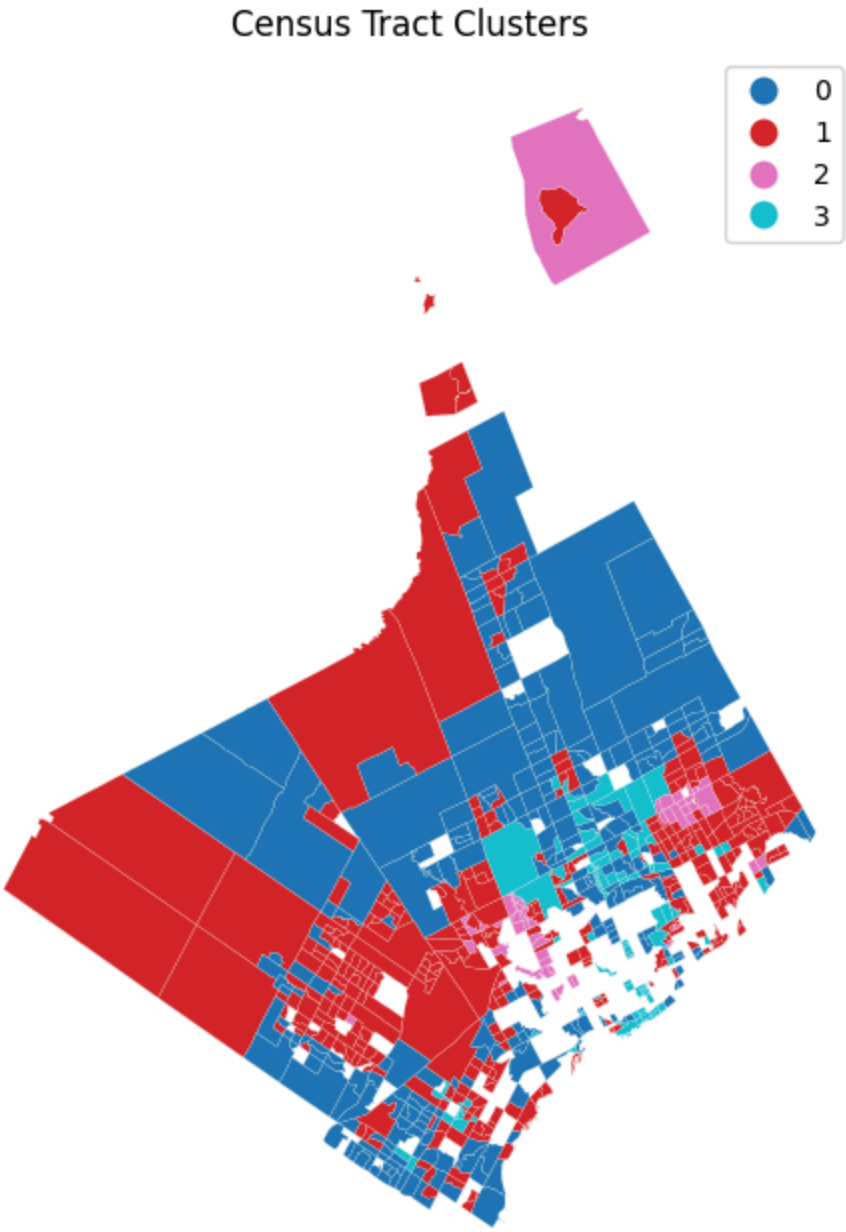


Figure 11: Cluster profile for Ward's hierarchical method n=4

cluster	0	1
Diabetes Percentage	5.78	8.91
Households by Income (Current Year) | Median Ho...	142474.03	91725.75
No certificate %	5.00	11.28
High (secondary) school %	17.32	24.92
Postsecondary Education %	77.66	63.78

cluster	2	3
Diabetes Percentage	10.48	5.05
Households by Income (Current Year) | Median Ho...	75027.95	82899.14
No certificate %	23.65	3.79
High (secondary) school %	29.51	13.47
Postsecondary Education %	47.10	82.75